Emission–OH partitioning in Qu et al. (2022)

**Reference.** Qu et al., *Environmental Research Letters* 17, 094003, doi:10.1088/1748-9326/ac8754. The full article and three-page supplement (Table S1, Figs. S1–S2) were read. The inversion covers 2019–2020 only; 2021–2022 remain missing.

**Method.** Independent annual analytical Bayesian inversions assimilate University of Leicester GOSAT v9 land and ocean-glint retrievals using GEOS–Chem 12.5.0, MERRA-2 meteorology, and a $2^\circ \times 2.5^\circ$ grid. The 4189-element state comprises 4020 annual non-wetland fluxes, 168 monthly wetland fluxes across 14 regions, and one annual global OH loss-frequency factor. Identical priors and prior covariances are used in both years: GFEI v2 fossil emissions, EDGAR v4.3.2 other anthropogenic emissions, and nine WetCHARTs v1.3.2 members for 2019 wetlands/lakes. Initial methane is independently scaled by 0.97 in 2019 and 1.01 in 2020 to match GOSAT. The base uncertainties are 50% for emissions and 10% for OH, with regularization $\gamma = 0.5$. Eight one-at-a-time sensitivity runs vary these errors and regularization.

**Native budget.** Rounded Table 1 gives burden 5197/5238 Tg, emissions 572/603, OH removal 469/468, and other removal 75/76 Tg $CH_4$ $yr^{-1}$ in 2019/2020. Thus

$$F_{\mathrm{OH},2020} = 469 - 468(5197/5238) = 4.66 \text{ Tg } CH_4 \text{ yr}^{-1}. \tag{1}$$

This agrees to reporting precision with the stated OH forcing of 5 and emission forcing of 31, totaling 36 Tg $CH_4$ $yr^{-1}$ (86% emissions). Ensemble means are $30 \pm 5.5$ for emissions and $82 \pm 18\%$ for their forcing share. Table S1 spans emission changes of 19–36 and total forcings of 35–39: the split is sensitive even when total forcing is comparatively stable. The base source–sink posterior error correlation is 0.97.

**Regional findings.** Africa contributes 15 Tg $CH_4$ $yr^{-1}$, 48% of the base emission increase. Canada/Alaska increases by 4.8, or 24% of its own 2019 emissions (about 15% of the global increase). China, CONUS, and South America decrease. Rainfall/inundation and boreal wetland temperature sensitivity are proposed explanations. The authors explicitly cannot robustly separate anthropogenic and wetland changes at grid-cell scale; the wetland interpretation uses ancillary evidence.

**Comparison limits.** The tabulated native growth anomaly is 31 Tg $CH_4$ $yr^{-1}$, compared with 14.24 from our NOAA snapshot. Annual initialization and different sampling/diagnostics must therefore be retained; replacing one baseline with the other would change attribution. Ratios of rounded burden and removal do not exactly equal the separate lifetime column, so we use one internally consistent pair of inputs. Other removal combines chemical and soil sinks. The data-availability DOI identifies GOSAT retrievals, not a posterior flux archive. Tables 1–2 and S1 are transcribed without attempting an inversion.